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# DiagPCNN: Enhancing CNN through Pretrained Model for Brain Tumor Diagnosis

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## Abstract

Analysis of medical images plays a vital role in the early detection and prognosis of diseases. Among various medical imaging, Magnetic Resonance Imaging(MRI) is valuable in providing insights into the brain artefact. Early classification of brain tumors has become indispensable for doctors and medical persons. With the recent advancements in deep learning, Convolution Neural Networks have gained widespread in the domain of disease diagnosis and medical imaging, due to their notable efficacy and high performance. In this study, we have proposed the DiagPCNN model which incorporates a pre-trained model with CNN for an increment in optimal efficacy of the manifest model. We have also analysed the pros and cons of the DiagPCNN model over eighteen different architectures and three DiagPCNN models outperforms the rest of models with superior accuracy of 96.63, 97.28 and 97.79% respectively. Hence, simplifying their skilfulness in detecting brain tumors to induce the manifest model of a productive and virtuous approach for enhanced prognosis.

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**Keywords:** Deep Learning, CNN; DiagPCNN; Brain MRI; Medical Imaging

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## 1. Introduction

Cells are the basic building and fundamental blocks of the human body. The cells continue to divide and grow forming new cells and the old cells present begin to die. When the cell loses its function and ability to grow normally and new cells grow abnormally along with the old cells present in the body, there arises a situation known as a tumor. A tumor is the abnormal growth of new cells in the part of the body when there are no requirements for it, forming a mass of lump or tissues. When this tumor does not spread to the rest part of the affected areas, then it is known as Benign cells which are non-cancerous. Meanwhile, if they spread locally to the affected regions, then it cancerous and henceforth it is known as Malignant cells. When this tumor begins to develop around the brain cells, then it is known as Brain Tumor [1-2]. The brain, the control center of the body is the focal and sensitive part of the nervous system. Brain cells are characterized by their softness and sponginess nature, which are non-replaceable. The main causes of brain tumors are inherited genetic predisposition, acquired genetic mutations, exposition to ionizing radiation, immune system disorders such as HIV and AIDS, increasing risk for a brain tumor with age, ethnicity, family history of brain tumor and There also exist some unknown causes. There are three types of brain tumors and they are Gliomas, Meningiomas and Pituitary Brain Tumor. Gliomas are the most common type of brain tumor that arises from the glial cells located within the brain [3]. The second type of brain tumor, Meningiomas originates from the preventive profane

between the spinal cord and the brain, known as meninges [4]. Headache, changes in vision, seizures, hearing loss and cognitive changes serve as the common symptoms of Gliomas and Meningiomas. The third and the last categorised form of brain tumor, the Pituitary tumor initiated from a small gland present at the base of the brain whose primary function is to regulate hormones, popularly known as the pituitary gland. The symptoms of a Pituitary tumor include headaches, visual sense difficulty, hormonal imbalance, change in energy level and weight and growth abnormalities. Brain tumors are the most crucial life-threatening state that requires an early diagnosis and prognosis for a functioning cure [5-6]. The detection and classification of brain tumors are manually examined and analyzed using Magnetic Resonance Imaging (MRI) by medical experts or doctors, which is time-consuming and also prone to human error, which leads to inefficient diagnosing of the disease [7].

Recent developments in the field of machine learning (ML) have paved a new path for medical image analysis. ML, basically a subset of artificial intelligence (AI), generally involves the model training to learn from the initial fed data and make predictions and decisions based on it [8-9]. Basic ML approaches viz, decision tree (DT) and support vector machine (SVM) can be used for medical image analysis but they require the manual extraction and selection of the features [10-11]. Deep learning (DL) approaches such as Convolutional Neural Networks (CNN) have emerged as the most powerful tools and the most efficient in the field of medical image analysis, as they can identify complex patterns and can learn hierarchically from the medical images [12]. CNN is a deep neural model that is broadly used in the field of medical imaging due to the explicit features of learning the data hierarchical and extracting the features spatially. Besides this, some other pre-trained models such as DenseNet [25], EfficientNet [26], ResNet [28], InceptionV3 [27], Xception [30], NASNet [29] and VGG [31] models are predominating in the field image classifications due to their optimization efficacy, architecture design and high performance. Therefore, the research aims to integrate these pre-trained model networks with the CNN model to increase their efficacy and to come out with a better diagnosis, classification and prognosis of the brain tumor.

The successive part of the study has been divided into the following sections. Section 2 outlines the state-of-the-art related to CNN and dense models for the detection of brain tumor. Section 3 represents the basic preliminaries required for the study. The manifest model setup has been determined in Section 4. Section 5 illustrates the result and interpretation of the proposed state-of-the-work. Lastly, the conclusion and subsequent reach are depicted in Section 6.

## 2. Related Works

Previous articles on brain tumor detection generally suffer from use the of limited standardised images and hence resulted in less reproducibility of the results. Older imaging methods might not be efficient in extracting some important features in the dataset. Hence, we outlined the analysis of the diversified author in optimize.

Akter et al. proposed a CNN model for the classification and segmentation of brain tumor using VGG16, VGG19, EfficientNetB0, EfficientNetB7 and Resnet152v2 evaluated on 6 Brain Magnetic Resonance Imaging (MRI) Dataset (2 merged and 4 augmented) and resulted that the model showed the best performance with the merged dataset 1 [13]. Srinivasan et al. proposed a hybrid model for brain tumor classification using 3-CNN approaches along with GoogleNet, AlexNet, DenseNet121, ResNet101 and VGG-16 on 3 datasets namely Figshare, Repository of Molecular Brain Neoplasia Data (REMBRANDT) and The Cancer Genome Atlas Low-Grade Glioma (TCGA-LGG) and concluded that the CNN approach-1 gave the best evaluative measures [14]. Suryawanshi and Patil proposed a hybrid model for Brain Tumor classification using CNN, VGG19 and CNN-SVM on the dataset of Brain Tumor Image Segmentation Benchmark (Brats) 2018 and Sartaj and concluded that CNN-SVM showed the best performance with both of the datasets [15]. Islam et al. proposed federated learning (FL) and CNN Ensemble model for brain tumor identification using VGG16, VGG19, InceptionV3, ResNet50, DenseNet121, and Xception on the dataset of Brain MRI and concluded that average CNN consisting of DenseNet121, VGG19, and InceptionV3 outperformed rest approaches [16]. Kumar proposed an analysis for segmentation of brain tumors using Otsu's, watershed, level set, K-means, HAAR Discrete Wavelet Transform (DWT), and CNN on the Brats 2018 dataset and concluded that the CNN model achieved the best performance among the rest of the models [17]. Patil and Kirange proposed an ensemble model for brain tumor detection using Shallow CNN (SCNN), VGG16 and Ensemble Deep Learning Model (EDCNN), an integration of SCNN and VGG16 on the Brain MRI dataset and concluded that EDCNN gave the highest measures among the models [18]. Rahman and Islam proposed a model integrating CNN and parallel deep convolutional neural network (PDCNN) for the detection and classification of a brain tumor on 3 datasets (Binary tumor identification, Figshare and Multiclass Kaggle dataset) and concluded that the model gave the best performance with the third dataset [19]. Saeedi et al. proposed a model for brain tumor detection using 2D CNN and Convolutional auto-encoder on the Brain MRI dataset and concluded that 2D CNN gave the best accuracy compared to Convolutional auto-encoder [20]. Zoghby et al. proposed a Dual Convolution Tumor Network, integrating

VGG16 with custom CNN on the Brats dataset and concluded that the model gave the best performance in contrast to another model [21]. Chattopadhyay and Maitra proposed a model for Brain tumor classification using CNN in Brats 2020 dataset and concluded that the model gave the highest accuracy compared to another model [22]. The sketch of state-of-the-work has been summarized in Table 1.

**Table 1.** Sketch of the state-of-the-art

| Reference                     | Methodology Employed                                                                                                        | Dataset Employed                                                                               | Year |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------|
| Akter et al. [13]             | VGG16, VGG19, EfficientNetB0, EfficientNetB7 and Resnet152v2                                                                | 6 Brain MRI Dataset                                                                            | 2024 |
| Srinivasan et al. [14]        | 3-CNN approaches, GoogleNet, AlexNet, DenseNet121, ResNet101 and VGG-16                                                     | 2 Merged Dataset and 4 Augmented Dataset, Segmentation Dataset Figshare, EMBRANDT and TCGA-LGG | 2024 |
| Suryawanshi and Patil [15]    | CNN, VGG19 and CNN-SVM                                                                                                      | Brats 2018 and Sartaj                                                                          | 2024 |
| Islam et al. [16]             | CNN (VGG16, VGG19, InceptionV3, ResNet50, DenseNet121, and Xception), Average CNN, Voting Ensemble, Federated Learning (FL) | Brain MRI                                                                                      | 2023 |
| Kumar [17]                    | Otsu's, watershed, level set, K-means, DWT and CNN                                                                          | Brats 2018                                                                                     | 2023 |
| Patil and Kirange [18]        | SCNN, VGG16 and EDCNN                                                                                                       | Brain MRI                                                                                      | 2023 |
| Rahman and Islam[19]          | CNN + PDCNN                                                                                                                 | Binary tumor identification, Figshare and Multiclass Kaggle dataset                            | 2023 |
| Saeedi et al. [20]            | 2D CNN, Convolutional auto-encoder                                                                                          | Brain MRI                                                                                      | 2023 |
| Zoghby et al. [21]            | Dual Convolution Tumor Network (VGG16 + Custom CNN)                                                                         | Brats                                                                                          | 2023 |
| Chattopadhyay and Maitra [22] | CNN                                                                                                                         | Brats 2020                                                                                     | 2022 |

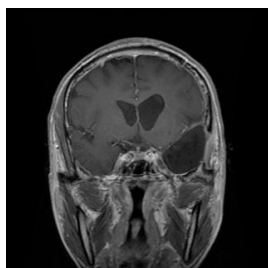
### 3. Basic Preliminaries

#### 3.1 Dataset Employed

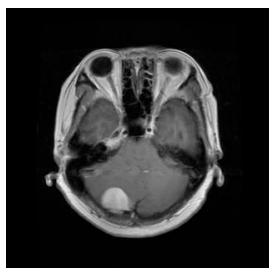
The article employs a Brain MRI dataset sourced from KAGGLE [23]. The dataset is a combination of Figshare, SARTAJ and Br35H datasets. The dataset consists of multi-label classes mainly Glioma Tumor, Meningioma Tumor, No Tumor and Pituitary Tumor, as depicted in Fig. 1 (a)-(d).. It is divided into training and testing sets, from which Glioma Tumor contains 1321 and 300, Meningioma Tumor contains 1339 and 306, No Tumor contains 1595 and 405 and Pituitary Tumor contains 1457 and 300 training and testing MRIs respectively. The image size incorporated in this dataset is 512 X 512. To ensure uniformity in training the neural network, all the castoff images are resized into 224 X 224. The resizing is done to ensure consistently, using the ImageDataGenerator present in the keras model. The description of the castoff dataset is depicted in Table 2.

**Table 2.** Dataset Description [23]

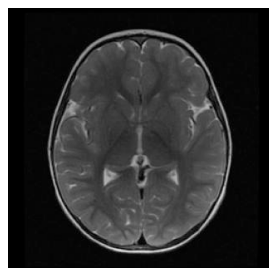
| Class Label | Class Name | Training Set | Testing Set |
|-------------|------------|--------------|-------------|
| 0           | Glioma     | 1321         | 300         |
| 1           | Meningioma | 1339         | 306         |
| 2           | No         | 1595         | 405         |
| 3           | Pituitary  | 1457         | 300         |



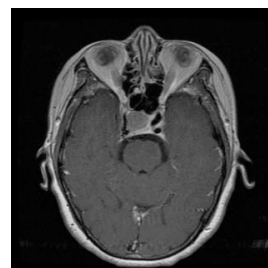
(a) Glioma



(b) Meningioma



(c) No



(d) Pituitary

**Fig. 1.** Brain MRI scans classifying the tumor [23]

### 3.2 CNN

CNN a deep learning-based model is widely used in the domain of medical imaging for their enormous capabilities of feature extraction. It can detect anomalies and patterns in the Brain MRI optimizing the model diagnosis conditions. It contains several layers such as a convolutional layer to filter the pre-processed input images, a pooling layer that typically helps in dimensionality reduction and fully connected dense layers that give the final output. This is the typical network of normal CNN. CNN can learn from hierarchical patterns and can handle high-dimensionality datasets very efficiently. Furthermore, for superior performance, the model also requires some significant computational assets [24]. The applications of CNN are not limited to, including text and image recognition, speech and video recognition, email spam detection and natural language processing (NLP). The basic working of the CNN model is depicted in Fig. 2.

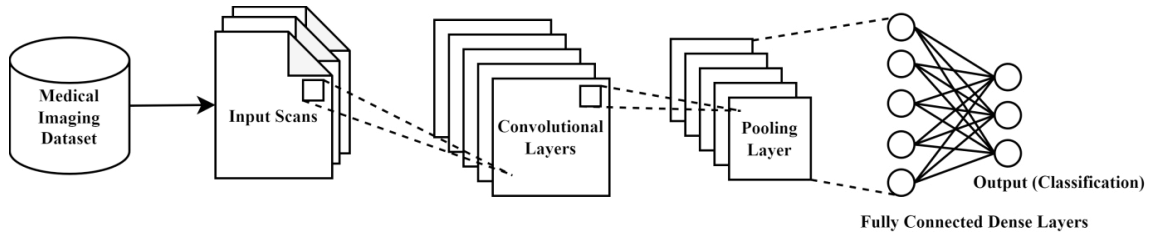


Fig. 2. Basic working of the CNN model

### 3.3 Pre-trained models

This work involves employing several kinds of pre-trained models including DenseNet121, DenseNet169, DenseNet201, EfficientNet(B0-B7), InceptionV3, ResNet50, NASNetMobile, NASNetLarge, Xception, VGG16 and VGG19 models incorporated with CNN to form the proposed hybrid model DiagPCNN for better Brain Tumor detection and classification. DenseNet models are deep learning networks that connect every layer present in a feed-forward style. This network optimizes the vanishing problem of gradient, enhancing the feature propagation and feature reuse. The number of parameters involved is also reduced. The number of layers in the model is generally referred to by the figures 121, 169 and 201. It is known for its best performance and efficient use mainly on the classification of images [25]. EfficientNet model is based on compound scaling. The size and complexity grew from model B0 to B7. It is a balanced model having proper resolution with depth and width. It can perform high on fewer parameters on image classification tasks [26]. InceptionV3 models use different sizes of multiple convolutional filters to handle the diversified features. It is widely used in this domain due to the integration of several methods like factorized convolutions and auxiliary classifiers. It consists of 48 layers of architecture making it popular for image classification tasks [27]. As the name suggests ResNet50, the model consists of a 50-layered residual deep network. It uses a skip connection which helps in the problem of vanishing gradient. It is a balanced model in both computational assets and complexity, making it more flexible in real-world applications for better image classification tasks [28].

The neural Architecture Search (NAS) method is used to automatically design NASNetMobile and NASNetLarge models. NASNetMobile requires less computational load with fewer parameters while NASNetLarge requires high computation efficiency and hence are strong performance model [29]. Xception are Extreme Inception model that is based upon depth-wise distinct convolutional networks. The model contains 36 convolutional layers, which are very organized for feature extraction. It is also a well-balanced model of high performance and computational resources [30]. The VGG model is known for their depth, simplicity and straightforwardness. The VGG16 model has 16 layers. Similarly, VGG19 has 19 layers. It typically uses a smaller filter size of 3x3 and generally has uniformness in its architecture distribution. In spite being requiring high computation resources, it can be widely used in image classification due to its easy implementation and understanding [31-32].

## 4. Proposed Method

The flow of the proposed model determines the succeeding (i) Identifying and Collecting the Brain MRI Dataset from KAGGLE (ii) Re-scaling of the raw inputted image dataset into (224,224) (iii) Splitting the cast-off dataset into an 80:20 ratio for train and test (iv) Building of the pre-trained models (v) Fitting of the pre-trained model to the CNN Model Architecture (vi) Employing novel hybrid model and comparing it with the basic pre-trained model and (vii) Persistency of the proposed hybrid work. The code for the proposed DiagPCNN model is made to run on the platform of Visual Studio Code on a Python environment using a GPU-based system having a CPU configuration of 8 GB of RAM and 4 GB of GTX 1650. A visual representation of the proposed hybrid model is depicted in Fig. 3.

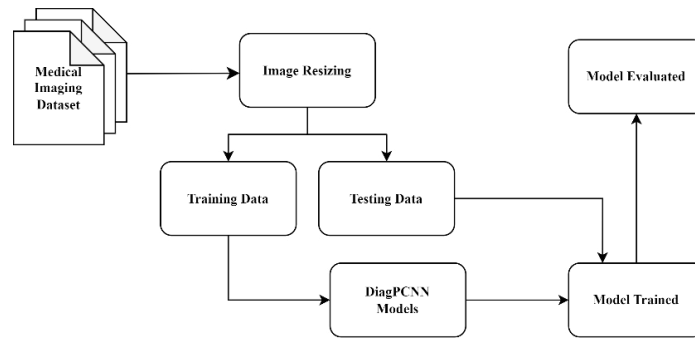


Fig. 3. Proposed Method

The proposed hybrid approach evaluates its model using the brain MRI dataset. The model architecture begins with the loading of the pre-trained model with an input shape of (224,224,3) for standard RGB images. Then the pre-trained layer of the model is set to False which prevents it from being updated during the model training. It is very useful in transfer learning. Followed by building a sequential model with the loaded pre-trained model at the base, preceded by adding two convolutional layers of 1024 and 512 filters with a kernel size of 3 x 3 and relu is used as the activation function. The BatchNormalization layer is added which increases the efficacy of the model by normalizing the activation used in the previous layers, followed by which GlobalAveragePooling2D layer, which is used to reduce the dimensionality and furnishes it for the approaching dense layers. Fully connected dense layers with 256 and 128 units with relu activation function are added with two dropout layers, which follow a regularization method that prevents overfitting of the model. In the dropout layer, almost half of the input batch is randomly set to zero. At last, a dense layer with 4 units (as there are 4 classes) is employed with an activation function of softmax, which converts the multi-class output to a probability distribution.

## 5. Result and Discussion

The performative metrics that evaluate the proposed hybrid DiagPCNN are as conferred in Eq. 1 to Eq. 4. Metrics including accuracy (*Acr*), precision (*Prn*), recall (*Rcl*), f1-score (*FSr*) and ROC curve with AUC value are evaluated on the Brain MRI dataset [33]. The instances supported by the model's execution are as presented in the Confusion Matrix (CM). It consists of four values namely true positive (*TP*): positive set predicted accurately, false positive (*FP*): positive set predicted inaccurately, true negative (*TN*): negative set predicted accurately and false negative (*FN*): negative set predicted inaccurately. The evaluative power of the model on the new test dataset is known as Precision. The higher value of precision is the indicator that the model is predicting more accurately. Recall is the ratio of the positive set predicted accurately to the total positive set. Handling the positive set efficiently indicates higher values of recall. F1-Score is composed of precision and recall. The model efficacy is also evaluated using the ROC curve. The ROC curve is a plot of true positive rate against the false positive rate. The area under the curve (AUC) has been computed to quantify the model's predictive power to classify the classes.

From the above study and experimentation, we conclude that pre-trained models viz., EfficientNetB1, ResNet50, and VGG16 incorporated with the CNN model achieved superior efficacy metrics obtained the accuracy of 96.63, 97.28 and 97.79% respectively as shown in Table 3 and 4. The CM along with the ROC curve for the above three models are presented in (a)-(f) of Fig. 6. EfficientNetB1 maintains the computational efficacy of the model due to the quality of compound scaling, which helps the network to balance its width, network, and resolution. ResNet50 can efficiently manage complex features due to its framework of deep residual learning. The simplified and straightforward network of VGG16 highlighted with the robust nature of feature extraction and computationally intensive, builds the best three superior models in the study. The model is trained for 5 Epoch, accuracy, and validation accuracy seem to be increasing in every epoch and the loss and validation loss seem to be decreasing in every epoch. The proposed hybrid DiagPCNN performance are superlative in contrast with that of pre-trained model. The accuracy attained from DiagPCNN is compared with contemporary models available in the literature, presented in Table 5. From the reported outcome, it's evident that our proposed DiagPCNN model show superiority over others. The graphical representation of efficacy metrics is depicted in Fig. 4.

$$Acr = \frac{TP + FP}{TP + FP + TN + FN} \quad (1)$$

$$Prn = \frac{TP}{TP + FP} \quad (2)$$

$$Rcl = \frac{TP}{TP + FP} \quad (3)$$

$$FSr = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (4)$$

**Table 3.** Efficacy metrics obtained over castoff hybrid approach

| DiagPCNN Model        | Accuracy     |              |              | Loss        |              |              |
|-----------------------|--------------|--------------|--------------|-------------|--------------|--------------|
|                       | Training     | Validation   | Testing      | Training    | Validation   | Testing      |
| DenseNet121           | 90.26        | 86.73        | 88.59        | 27.15       | 37.41        | 34.13        |
| DenseNet169           | 89.54        | 85.97        | 82.88        | 23.89       | 31.66        | 36.85        |
| DenseNet201           | 91.41        | 84.54        | 84.03        | 22.10       | 37.51        | 39.65        |
| EfficientNetB0        | 97.04        | 94.17        | 91.25        | 9.38        | 14.47        | 21.95        |
| <b>EfficientNetB1</b> | <b>96.63</b> | <b>94.37</b> | <b>94.29</b> | <b>8.48</b> | <b>12.92</b> | <b>18.54</b> |
| EfficientNetB2        | 95.13        | 93.79        | 93.15        | 12.37       | 16.01        | 16.95        |
| EfficientNetB3        | 96.58        | 94.56        | 92.01        | 9.36        | 14.59        | 20.20        |
| EfficientNetB4        | 97.11        | 94.27        | 92.39        | 8.29        | 16.06        | 19.03        |
| EfficientNetB5        | 96.63        | 93.03        | 92.39        | 9.31        | 17.43        | 24.49        |
| EfficientNetB6        | 95.18        | 92.84        | 90.11        | 12.67       | 19.26        | 26.43        |
| EfficientNetB7        | 95.78        | 93.51        | 92.01        | 12.27       | 20.92        | 21.81        |
| InceptionV3           | 83.84        | 78.62        | 79.46        | 41.37       | 58.35        | 59.67        |
| <b>ResNet50</b>       | <b>97.28</b> | <b>94.65</b> | <b>93.53</b> | <b>6.90</b> | <b>15.21</b> | <b>13.82</b> |
| NASNetMobile          | 90.61        | 83.77        | 92.50        | 26.48       | 42.76        | 43.39        |
| NASNetLarge           | 81.68        | 77.95        | 77.18        | 41.00       | 53.75        | 62.97        |
| Xception              | 90.89        | 85.20        | 80.60        | 23.14       | 40.27        | 41.16        |
| <b>VGG16</b>          | <b>97.79</b> | <b>94.94</b> | <b>95.81</b> | <b>6.10</b> | <b>15.26</b> | <b>16.91</b> |
| VGG19                 | 97.74        | 93.89        | 92.39        | 6.95        | 14.54        | 20.86        |

**Table 4.** Macro and Weighted Averaging metrics obtained over castoff hybrid approach

| DiagPCNN Model        | Macro Averaging |              |              | Weighted Averaging |              |              |
|-----------------------|-----------------|--------------|--------------|--------------------|--------------|--------------|
|                       | Precision       | Recall       | F1-Score     | Precision          | Recall       | F1-Score     |
| DenseNet121           | 88.75           | 87.66        | 87.33        | 89.07              | 88.59        | 88.03        |
| DenseNet169           | 87.72           | 82.02        | 81.22        | 88.48              | 82.89        | 82.31        |
| DenseNet201           | 85.42           | 82.66        | 82.79        | 85.28              | 84.03        | 83.43        |
| EfficientNetB0        | 91.47           | 90.77        | 90.91        | 91.99              | 91.25        | 91.43        |
| <b>EfficientNetB1</b> | <b>94.00</b>    | <b>93.80</b> | <b>93.85</b> | <b>94.38</b>       | <b>94.30</b> | <b>94.29</b> |
| EfficientNetB2        | 93.08           | 92.64        | 92.67        | 93.27              | 93.16        | 93.04        |
| EfficientNetB3        | 91.98           | 91.36        | 91.19        | 92.52              | 92.02        | 91.82        |
| EfficientNetB4        | 92.13           | 91.74        | 91.83        | 92.41              | 92.40        | 92.30        |
| EfficientNetB5        | 92.53           | 91.74        | 91.95        | 92.55              | 92.40        | 92.29        |
| EfficientNetB6        | 91.51           | 89.54        | 89.54        | 92.14              | 90.11        | 90.22        |
| EfficientNetB7        | 92.03           | 91.29        | 91.29        | 92.16              | 92.02        | 91.74        |
| InceptionV3           | 79.52           | 77.59        | 75.14        | 79.96              | 79.47        | 76.55        |
| <b>ResNet50</b>       | <b>93.39</b>    | <b>92.94</b> | <b>92.86</b> | <b>93.79</b>       | <b>93.54</b> | <b>93.38</b> |
| NASNetMobile          | 82.55           | 81.07        | 80.55        | 83.35              | 82.51        | 81.75        |
| NASNetLarge           | 84.79           | 75.58        | 71.74        | 85.61              | 77.19        | 73.58        |
| Xception              | 79.85           | 79.01        | 78.62        | 80.04              | 80.61        | 79.52        |
| <b>VGG16</b>          | <b>95.87</b>    | <b>95.48</b> | <b>95.51</b> | <b>96.11</b>       | <b>95.82</b> | <b>95.81</b> |
| VGG19                 | 91.98           | 91.73        | 91.80        | 92.26              | 92.40        | 92.27        |

**Table 5.** Performance assessment of DiagPCNN over state-of-the-art approaches

| Model                           | Accuracy     |
|---------------------------------|--------------|
| <b>EfficientNetB1(Proposed)</b> | <b>96.63</b> |
| <b>ResNet50(Proposed)</b>       | <b>97.28</b> |
| <b>VGG16(Proposed)</b>          | <b>97.79</b> |
| Akter et al. [13]               | 89.4         |
| Islam et al. [16]               | 91.05        |
| Sacedi et al. [20]              | 95.63        |
| Islam et al. [16]               | 96.68        |
| Sacedi et al. [20]              | 96.47        |

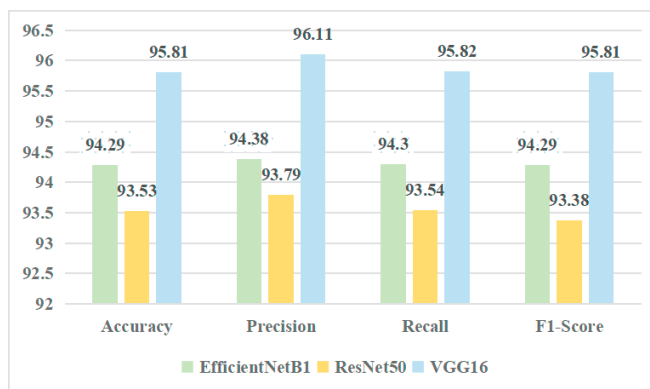


Fig. 4. Performance evaluation graph relating to the parameters

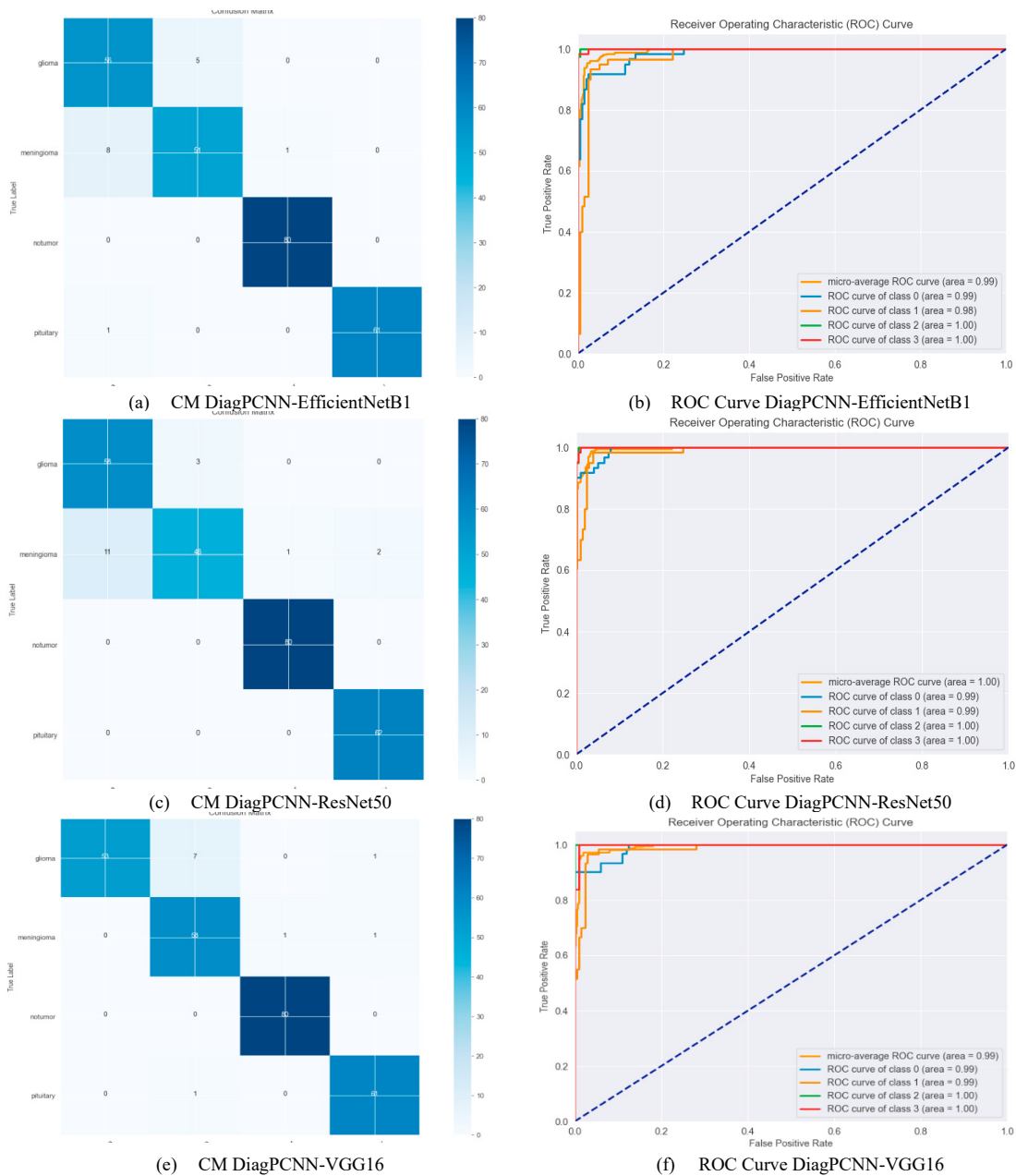


Fig. 5. Aftermath obtained over evaluating the proposed DiagPCNN model using Brain MRI Dataset

## 6. Conclusion and Future Scope

This study emphasizes the application of a trained dense model integrated with CNN for the classification and diagnosis of Brain Tumors. Among 18 different architectures, CNN incorporated with EfficientNetB1, ResNet50 and VGG16 have outperformed all the models with superior evaluation metrics, achieving the model's highest accuracy of 96.63, 97.28 and 97.79% respectively, making the proposed DiagPCNN the most efficient analysis models this context. While EfficientNetB1 is known for its compound scaling method holding back the computation cost, ResNet50 retains its ability of residual learning to handle the complex feature and VGG16 due to its depth, simplicity and robust nature for feature extraction makes the proposed model more efficient in classifying the Brain MRI.

In future context, the CNN model can be optimized using fine-tuning, transfer learning and hyperparameter tuning. Furthermore, the model can be evaluated with some real-time dataset exploring the susceptibility of the model. In addition to this, a bio-inspired contemporary meta-heuristic optimization approach can also be integrated with the model to deduce the efficacy metrics of the proposed hybrid state-of-the-art.

## References

1. Batool, A. and Byun, Y.C., 2024. Brain tumor detection with integrating traditional and computational intelligence approaches across diverse imaging modalities-Challenges and future directions. *Computers in Biology and Medicine*, p.108412.
2. Kienzler, J.C. and Becher, B., 2024. Immunity in malignant brain tumors: Tumor entities, role of immunotherapy, and specific contribution of myeloid cells to the brain tumor microenvironment. *European Journal of Immunology*, 54(2), p.2250257.
3. Mokri, S.M.G., Valadbeygi, N. and Grigoryeva, V., 2024. Diagnosis of Glioma, Meningioma and Pituitary brain tumor using MRI images recognition by Deep learning in Python. *EAI Endorsed Transactions on Intelligent Systems and Machine Learning Applications*, 1.
4. Watanabe, G., Young, K., Rauber, E., Khan, M.F., Suzuki, R., Riestenberg, R., Umana, G.E. and Palmisciano, P., 2024. A systematic review of extracranial meningioma metastasis: timing, evolution and outlook. *Journal of Neuro-Oncology*, pp.1-10.
5. Sajjad, G., Khan, M.B.S., Ghazal, T.M., Saleem, M., Khan, M.F. and Wannous, M., 2023, March. An Early Diagnosis of Brain Tumor Using Fused Transfer Learning. In *2023 International Conference on Business Analytics for Technology and Security (ICBATS)* (pp. 1-5). IEEE.
6. Warjurkar, S.V. and Riddhorkar, S., 2024. Maximizing Precision in Early Prognosis using SVM-ACO Classifier and Hybrid Optimization Techniques in MRI Brain Tumor Segmentation with Integration of Multi-Modal Imaging Data. *International Journal of Intelligent Systems and Applications in Engineering*, 12(10s), pp.389-401.
7. Aggarwal, M., Tiwari, A.K., Sarathi, M.P. and Bijalwan, A., 2023. An early detection and segmentation of Brain Tumor using Deep Neural Network. *BMC Medical Informatics and Decision Making*, 23(1), p.78.
8. Panda, N., Mohanty, P., Rao, G.N. and Tulsibabu, S., 2022, February. Application of Machine Learning Model Based Techniques for Prediction of Heart Diseases. In *International Conference on Metaheuristics in Software Engineering and its Application* (pp. 266-274). Cham: Springer International Publishing.
9. Dalai, A.K., Jena, A.K., Ramana, B.V., Maneesha, B. and Panda, N., 2022, February. Supervised machine learning approaches for medical data classification. In *2022 2nd International Conference on Artificial Intelligence and Signal Processing (AISP)* (pp. 1-8). IEEE.
10. Jagadeeswara Rao, G., Siva Prasad, A., Sai Srinivas, S., Sivaparthi, K. and Panda, N., 2022. Data classification by ensemble methods in machine learning. In *Advances in Intelligent Computing and Communication: Proceedings of ICAC 2021* (pp. 127-135). Singapore: Springer Nature Singapore.
11. Jena, A.K., Gopal, K.M., Tripathy, A. and Panda, N., 2023, January. Review Sentiment Classification and Feature Selection Using Hybridized Support Vector Machine. In *International Conference on Communication and Computational Technologies* (pp. 327-337). Singapore: Springer Nature Singapore.
12. Panda, N., Mohanty, P., Naidu, B.R. and Prasad, G.K., 2023, June. Application of deep convolutional neural network for facial appearance recognition. In *AIP Conference Proceedings* (Vol. 2819, No. 1). AIP Publishing.
13. Akter, A., Nosheen, N., Ahmed, S., Hossain, M., Yousuf, M.A., Almoyad, M.A.A., Hasan, K.F. and Moni, M.A., 2024. Robust clinical applicable CNN and U-Net based algorithm for MRI classification and segmentation for brain tumor. *Expert Systems with Applications*, 238, p.122347.
14. Srinivasan, S., Francis, D., Mathivanan, S.K., Rajadurai, H., Shivahare, B.D. and Shah, M.A., 2024. A hybrid deep CNN model for brain tumor image multi-classification. *BMC Medical Imaging*, 24(1), p.21.
15. Suryawanshi, S. and Patil, S.B., 2024. Efficient Brain Tumor Classification with a Hybrid CNN-SVM Approach in MRI. *Journal of Advances in Information Technology*, 15(3).
16. Islam, M., Reza, M.T., Kaosar, M. and Parvez, M.Z., 2023. Effectiveness of federated learning and CNN ensemble architectures for identifying brain tumors using MRI images. *Neural Processing Letters*, 55(4), pp.3779-3809.
17. Kumar, A., 2023. Study and analysis of different segmentation methods for brain tumor MRI application. *Multimedia Tools and Applications*, 82(5), pp.7117-7139.
18. Patil, S. and Kirange, D., 2023. Ensemble of deep learning models for brain tumor detection. *Procedia Computer Science*, 218, pp.2468-2479.
19. Rahman, T. and Islam, M.S., 2023. MRI brain tumor detection and classification using parallel deep convolutional neural networks. *Measurement: Sensors*, 26, p.100694.
20. Saeedi, S., Rezayi, S., Keshavarz, H. and R. Niakan Kalhori, S., 2023. MRI-based brain tumor detection using convolutional deep learning methods and chosen machine learning techniques. *BMC Medical Informatics and Decision Making*, 23(1), p.16.
21. Al-Zoghby, A.M., Al-Adawly, E.M.K., Moawad, A., Yehia, N. and Ebada, A.I., 2023. Dual deep CNN for tumor brain classification. *Diagnostics*, 13(12), p.2050.

22. Chattopadhyay, A. and Maitra, M., 2022. MRI-based brain Tumor image detection using CNN based deep learning method. *Neuroscience informatics*, 2(4), p.100060.
23. Nickparvar, M., 2021. Brain tumor MRI dataset. Kaggle.
24. Balamurugan, T. and Gnanamanoharan, E., 2023. Brain tumor segmentation and classification using hybrid deep CNN with LuNetClassifier. *Neural Computing and Applications*, 35(6), pp.4739-4753.
25. Vinisha, A. and Boda, R., 2024. Efficient brain Tumor detection system by Cascaded Fully Convolutional Improved DenseNet with Attention-based Adaptive Swin Unet-derived segmentation strategy. *Multimedia Systems*, 30(3), p.134.
26. Ghosh, A., Soni, B. and Baruah, U., 2024. A Fine-Tuned EfficientNet B1 Based Deep Transfer Learning Framework for Multiple Types of Brain Disorder Classification. *Iranian Journal of Science and Technology, Transactions of Electrical Engineering*, pp.1-21.
27. Bedi, P., Ningshen, N., Rani, S. and Gole, P., 2023, February. Explainable Predictions for Brain Tumor Diagnosis Using InceptionV3 CNN Architecture. In *International Conference On Innovative Computing And Communication* (pp. 125-134). Singapore: Springer Nature Singapore.
28. T. R. M., V. V.K. and Guluwadi, S., 2024. Enhancing brain tumor detection in MRI images through explainable AI using Grad-CAM with Resnet 50. *BMC Medical Imaging*, 24(1), p.107.
29. Altamimi, A., Alrowais, F., Karamti, H., Umer, M., Cascone, L. and Ashraf, I., 2024. An improved skin lesion detection solution using multi-step preprocessing features and NASNet transfer learning model. *Image and Vision Computing*, 144, p.104969.
30. Kaur, G., Sharma, N., Upadhyay, D., Singh, M. and Gupta, R., 2024, March. Brain Tumor Disease Detection in MRI Images using Xception Model. In *2024 3rd International Conference for Innovation in Technology (INOCON)* (pp. 1-5). IEEE.
31. Veni, N. and Manjula, J., 2023. VGG-16 architecture for MRI brain tumor image classification. In *Futuristic Communication and Network Technologies: Select Proceedings of VICFCNT 2021, Volume 1* (pp. 319-328). Singapore: Springer Nature Singapore.
32. Rao, R.S., Kalabarige, L.R., Holla, M.R. and Sahu, A.K., 2024. Multimodal Imputation based Multimodal autoencoder framework for AQI classification and prediction of Indian cities. *IEEE Access*.
33. Bendi, R.V., Panda, N., Teja, S., Mohapatra, H., Dalai, A. and Majhi, S.K., 2023. Improved Chaotic Grey Wolf Optimization for Training Neural Networks: IMPROVED CGWO FOR TRAINING NEURAL NETWORKS. *Journal of Scientific & Industrial Research (JSIR)*, 82(11), pp.1193-1207.